Homework 1

Monday, March 9, 2026 10:46 PM

1. $w^T = [1 \;\; 3]$

$w^T x = \underset{1\times 2}{[1 \;\; 3]} \underset{2\times 1}{\begin{bmatrix} 2 \\ -1 \end{bmatrix}} = 2-3=-1$

$z = -1-2=-3$

2. $z = 4(-0.5)+1 = -2+1 = -1$

$\hat{y} = \sigma(-1) = \frac{1}{1+e}$

3. $W_1 x = \underset{2\times 2}{\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} 1 \\ 2 \end{bmatrix}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$W_2 h = \underset{1\times 2}{[2 \;\; -1]} \underset{2\times 1}{\begin{bmatrix} 0 \\ 3 \end{bmatrix}} = -3$

$z_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$y = -3+0 = -3$

$h = \text{ReLU}\left(\begin{bmatrix} -1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

4. $L = \frac{1}{2}(-1-2)^2 = \frac{1}{2}(9) = \frac{9}{2}$

5. $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$

6. $\hat{y} = [0 \;\; 1]\begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1 = -2+1 = -1$

$(\hat{y}-y) = -1-0 = -1$

$\frac{\partial L}{\partial w_1} = (\hat{y}-y)(x_1) = -1(1) = -1$

$\frac{\partial L}{\partial w_2} = (\hat{y}-y)(x_2) = -1(-2) = 2$

$\frac{\partial L}{\partial b} = (\hat{y}-y)(1) = -1$

7. Updated $w = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 0.2\left(\begin{bmatrix} -1 \\ 2 \end{bmatrix}\right)$

$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.2 \\ -0.4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.6 \end{bmatrix}$

Updated $b = 1 - 0.2(-1) = 1+0.2 = 1.2$

8. $z = 2(-1) - 1 = -3$

$a = 0$

$\frac{\partial L}{\partial w} = (a-y)\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial w} = (a-y)\cdot\frac{\partial a}{\partial z}\cdot x = 0$

$\frac{\partial L}{\partial b} = (a-y)\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial b} = (a-y)\cdot\frac{\partial a}{\partial z}\cdot 1 = 0$

9. $\frac{\partial L}{\partial w} = (\hat{y}-y)\cdot\frac{\partial \hat{y}}{\partial w} = x\cdot(\hat{y}-y)$

$\frac{\partial L}{\partial b} = (\hat{y}-y)\cdot\frac{\partial \hat{y}}{\partial b} = \hat{y}-y$

$\hat{y}_0 = 0(2)+0 = 0$

$w_1 = 0-(0.1)(2(-1)) = 0 - \frac{1}{10}(-2) = \frac{1}{5}$

$b_1 = 0-(0.1)(-1) = \frac{1}{10}$

$\hat{y}_1 = \frac{1}{5}(2) + \frac{1}{10} = \frac{1}{2}$

$w_2 = \frac{1}{5} - (0.1)(-\frac{1}{2}\cdot 2) = \frac{1}{5} - \frac{1}{10}(-1) = \frac{1}{5}+\frac{1}{10} = \frac{3}{10}$

$b_2 = \frac{1}{10} - (0.1)(-\frac{1}{2}) = \frac{1}{10} - \frac{1}{10}(-\frac{1}{2}) = \frac{1}{10}+\frac{1}{20} = \frac{3}{20}$

10. $\text{sum} = e^0 + e^1 + e^2 = 1+e+e^2$

$\text{softmax}(s) = \begin{bmatrix} \frac{e^0}{\text{sum}} \\ \frac{e^1}{\text{sum}} \\ \frac{e^2}{\text{sum}} \end{bmatrix} = \begin{bmatrix} \frac{1}{1+e+e^2} \\ \frac{e}{1+e+e^2} \\ \frac{e^2}{1+e+e^2} \end{bmatrix}$

11. During the forward pass, the input goes through the hidden layers, leading to a predicted output. During the back propagation, we seek to minimize the cost function by calculating the gradients of the weights and biases and updating them using the chain rule as the calculations start from the last layer and move backwards.
12. If a neural network used no nonlinear activation functions, it wouldn't be able to train over non-linear patterns in the model, as it would just behave like a singular layer.
13. If the learning rate is too large, the model can over-shoot, taking too large steps and skipping over the minimum of the loss function. But if the learning rate is too small, it will take a long time for the model to learn and it might get stuck in the local minimum.
14. Vanishing gradients are when the gradients get smaller, vanishing as the inputs are large or small, negatively affecting the learning process as the gradient will be cut off and the weights will stop updating.
15. ReLU is preferred over sigmoid because it is easier to train and more stable, as its derivatives are well behaved, preventing vanishing gradients and it's a simple nonlinear transformation.